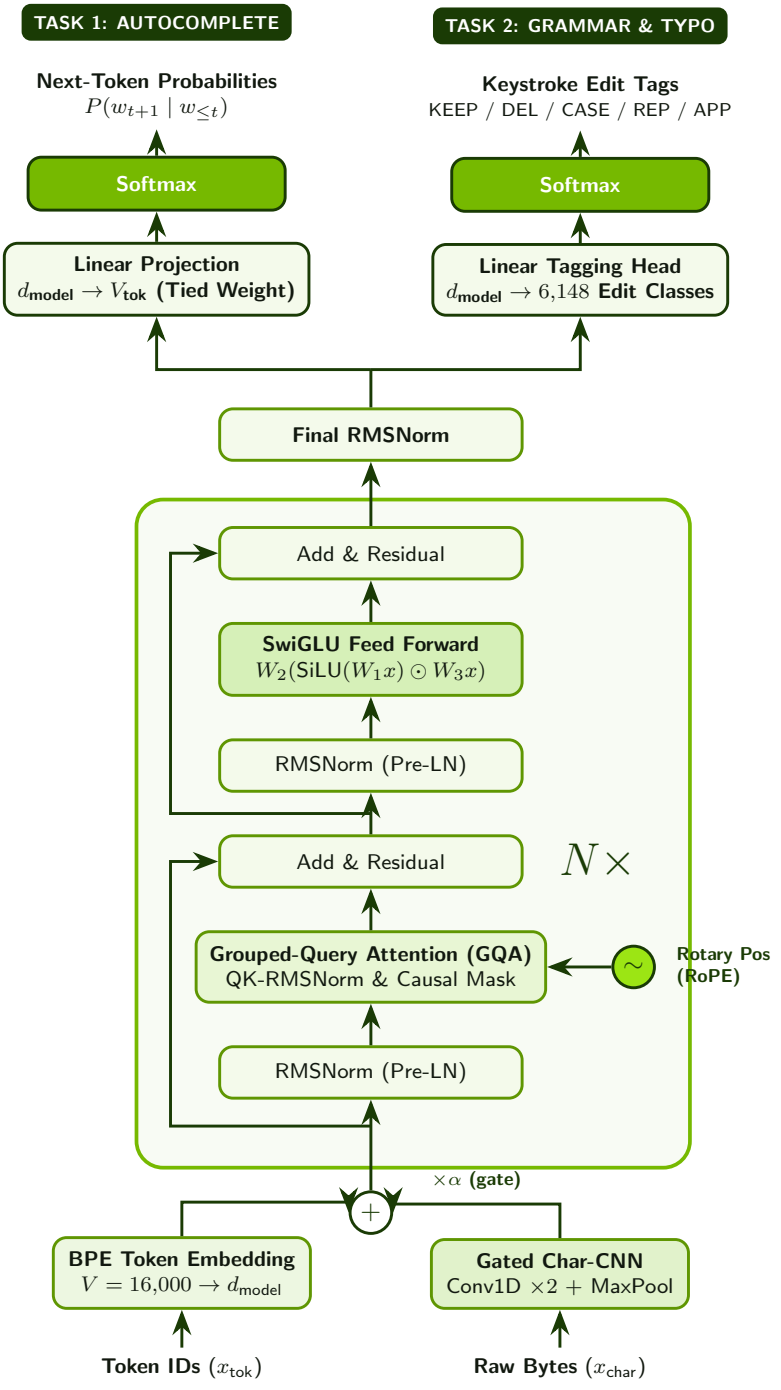

TASK 1: AUTOCOMPLETE
TASK 2: GRAMMAR & TYPO
Next-Token Probabilities
$P(w_{t+1} \mid w_{\leq t})$
Keystroke Edit Tags
KEEP / DEL / CASE / REP / APP
Softmax
Softmax
Linear Projection
$d_{\text{model}} \rightarrow V_{\text{tok}}$ (Tied Weight)
Linear Tagging Head
$d_{\text{model}} \rightarrow 6{,}148$ Edit Classes
Final RMSNorm
Add & Residual
SwiGLU Feed Forward
$W_2(\text{SiLU}(W_1 x) \odot W_3 x)$
RMSNorm (Pre-LN)
Add & Residual
$N\times$
Grouped-Query Attention (GQA)
QK-RMSNorm & Causal Mask
Rotary Pos (RoPE)
RMSNorm (Pre-LN)
$\times\alpha$ (gate)
BPE Token Embedding
$V = 16{,}000 \rightarrow d_{\text{model}}$
Gated Char-CNN
Conv1D $\times 2$ + MaxPool
Token IDs (x_{tok})
Raw Bytes (x_{char})